

BioFauna: a frozen BioCLIP-2.5 ViT-H retrieval system for Mediterranean marine identification

Author: Gustavo Zafra (Yespi)

Taxonomic contributors: Xavier Salvador, Miquel Pontes, Manuel Ballesteros

Repository: <https://github.com/yespi/biofauna>

Live system: <https://fotofauna.yespi.es>

This version: 2026-09-14 (frozen snapshot of production)

> **Naming.** Developed as YOLOFauna (2024–mid-2026); renamed BioFauna when production settled on BioCLIP retrieval rather than YOLO detection. Provenance: `docs/HISTORY.md`.

Update 2026-09-28 — gallery hygiene, an honest field evaluation, and a zero-shot rescue. After the leak fix (2026-09-23) the closed-set figure dropped to 88.11% on the leak-free set; the *operating* evaluation moved to a **field evaluation** assembled only from field photos (iNaturalist/Minka, research grade, 20–30 photos per species, leak-audited before merging): **62,408 rows** → **81.4% species / 84.9% genus**. Gallery hygiene in the same period: **135 duplicate synonym slugs merged** (name authority: **Minka**; if both names exist in Minka they are *not* merged; otherwise WoRMS synonymy resolves the equivalence), **~1,000 mislabelled photos relocated** (Minka re-identifications) with manifests and full rollback, and 57 photos quarantined. Two new production guards: (a) a **k-NN margin guard** in AutoID (a species is not published when the top-1/top-2 similarity margin is below 0.02 — those rows score 53% in the field evaluation); (b) an automatic **curator-correction guard** that retracts our identifications when a curator corrects at class level or above. A **catalogue-wide zero-shot second opinion** (BioCLIP-2.5 text embeddings; 2,985 labels) is now used as a *rescue*: when the k-NN is unsure (top-1 similarity < 0.85) and the zero-shot agrees with a different species at $p \geq 0.90$, the zero-shot answer wins — measured **+0.91 pp** on the field evaluation (fix 772 / break 315, 15% coverage) and now live. The temporal seasonal prior was **discarded** at full scale (76.84% → 76.64% best variant). Everything is audited: contaminación por prototipo (daily), leak QA of mined evaluation rows (weekly, with automatic purge).

Abstract

BioFauna identifies Mediterranean (and incidental adjacent) taxa from photographs by **retrieval** over a regional gallery, not by fine-tuning a new classifier. The production encoder is **BioCLIP-2.5 ViT-H/14** (632M parameters, 1024-d embeddings), **frozen**. Queries are encoded (with a 65% centre-crop fused into the global embedding), searched with **k-NN (k=15)** using a tempered vote aggregator, optionally re-weighted by a geographic prior, then mapped to calibrated probabilities and, when the margin is weak, **abstained** to genus, family, or a curated indistinguishability group.

Production snapshot (2026-09-28, live /health + calibration.json):

Quantity	Value
----- -----	
Gallery	1,118,353 embeddings / 4,543 species, FAISS row-aligned

Catalog (Mediterranean checklist)	2,985 taxa (dataset/catalog.json)
Out-of-sample accuracy (leak-free)	88.11% species / 90.55% genus / 92.46% family
Evaluation size	n= 12,373 observations, 2,091 species with ≥ 1 eval sample (gallery copies removed, O18)
Hardware	NVIDIA RTX 3060 12 GB (~4.4 GB VRAM at inference)

Those accuracy figures are **observation-stratified and leak-checked with a global-embedding gate** (the earlier 85.78% on n=19,087 and the later 92–93% panel contained gallery copies — see the update box and O18). They are **not** comparable to the August 2026 headline of 75.97% on a smaller, earlier cohort (n=12,788) without re-running that same harvest; mixing cohorts is how this project previously overstated progress. Both numbers are kept below, labelled by protocol.

Update 2026-09-23 — evaluation leak found and removed. An audit that embeds every evaluation photo *globally* (no TTA, exactly like the gallery) and compares it with its own species' gallery found that **30.0%** of the 25,143 evaluation rows were byte-level copies of a gallery photo (cosine ≥ 0.995) and **50.8%** were copies or rescaled/cropped copies (≥ 0.98). The harvest leak gate compared a *TTA-averaged* query embedding against a 0.999 threshold, which an identical image never reaches (~ 0.99), so it silently passed everything after the August fix (§3). Leaked rows scored **97.6%**; the remaining clean rows score **88.11%** species / 90.55% genus / 92.46% family (n=12,373, 2,091 species). The panel figure of 92.4–93.4% quoted on 2026-09-21/22 is therefore withdrawn; **88.1% is the current closed-set figure**, and the $\sim 80\%$ recent-observation KPI (O16) still stands as the operating number. For rare species with new, genuinely unseen photos from other sources (Wikimedia, GBIF, museum media), accuracy was only **24%** (n=130) — the leak had hidden it. Gate fixed in all harvesters, leaked rows moved aside (not deleted), calibrator refit on clean data (O18). Also in production since 2026-09-22: per-species cap of 3 votes in the k-NN aggregator (K23).

Update 2026-09-21. Production now serves **1,118,353** embeddings / **4,543** species (1,567 are gallery classes outside the 2,985-species catalog: 1.8% of vectors) after two growth waves (K19–K20). The panel (out-of-sample overlay, n=24,475) reads **92.36%** species; a check on 300 recent Minka research-grade observations gives $\sim 80\%$ (in-catalog birds 84% vs 96% on the panel) — the panel is a closed-set evaluation and overstates real-world accuracy (O16). Closed-set retrieval answers confidently for species missing from the catalog; an independent BioCLIP zero-shot second opinion removes most of those errors in birds (K21).

A systematic search for extra species top-1 from **training on this embedding space** (QLoRA, LoRA, triplet, ArcFace, linear head, scoped SupCon) **did not beat frozen retrieval**. Gains that shipped were backbone scale, gallery completeness/quality, inference-time fusion, a tempered k-NN aggregator, and taxonomic abstention. We publish the experiment ledger, taxon IDs, prototype centroids, and a self-hostable identifier. **Photographs and per-photo embeddings.npy are not redistributed**; they can be rebuilt from Minka / iNaturalist / GBIF using the catalog.

Keywords: BioCLIP-2.5, ViT-H, k-NN, fine-grained visual classification, marine biodiversity, citizen science, taxonomic abstention, calibration, Mediterranean Sea

1. Introduction

The Mediterranean holds a disproportionate share of marine biodiversity in a small basin, while taxonomic capacity is scarce. Citizen-science networks (Minka, iNaturalist) collect photographs faster than experts can identify them. Global classifiers help, but regional endemics, cryptic invertebrates, and language-specific field knowledge remain a bottleneck.

BioFauna is the identifier behind FotoFauna. Design choice: **do not train a closed-set CNN**. Use a biology-pretrained vision encoder and a **regional reference gallery** that can grow species by species without retraining.

Contributions

- 1. A production retrieval stack (frozen ViT-H + k-NN + calibration + hierarchical abstention) running on consumer GPU and publishing high-confidence IDs to Minka.
- 2. An **observation-stratified** evaluation protocol, including a gallery-similarity leak check after a 42.7% self-duplicate incident.
- 3. A **ledger of experiments** (this paper §4 and docs/EXPERIMENTS.md): each trial has a hypothesis, a result, and whether it was kept.
- 4. Open reconstruction artefacts: taxon IDs, prototype centroids ($4,543 \times 1024$), calibrators, geo priors, cryptic-pair lists — not the photo corpus.

2. System

2.1 Encoder

Property	Value
----- -----	
Checkpoint	hf-hub:imageomics/bioclip-2.5-vith14
Architecture	ViT-H/14, ~632M params, 1024-d L2-normalized embeddings
Input	224×224
Training in BioFauna	Frozen

2.2 Identification (production, 2026-09-23)

- 1. Embed the query; fuse **global frame + 65% centre crop** (ROI fusion; replaces an earlier 90% TTA).
- 2. Retrieve **k=15** gallery neighbours (cosine / inner product via FAISS when the full gallery is present; nearest-centroid over published prototypes otherwise).
- 3. Aggregate neighbour scores with **tempered votes** $\sum \exp(\max(s, 0)/T)$, **T=0.05**, plus a prototype-similarity boost. Each species contributes **at most 3** of the 15 neighbours to the vote (**KNN_CLASS_CAP=3**, K23), so a species with a huge gallery cannot outvote a sparse one by sheer count.
- 4. Optional **multiplicative geographic prior** when GPS is present.
- 5. **Abstain** when the top-1/top-2 margin is small and taxa share genus/family, or when the pair/genus/group is in **dataset/taxonomic_exceptions.json**.
- 6. Map k-NN features → **P(correct)** with logistic regression (**dataset/calibration.json**).

AutoID on FotoFauna publishes when calibrated **p ≥ 0.80**. With the calibrator refit on leak-free data (**created=2026-09-23T08:22**, fit 8,509 / test 3,864 rows, species-disjoint split) that threshold gives **97.0%**

precision at 80.6% coverage on the test split (closed-set; the recent-observation KPI is lower, O16). The August estimate (95.3% at 57.4%) is superseded.

This public repository ships **prototype centroids** (`data/patterns/<slug>/prototype.npy`). That is enough for a nearest-centroid demo. Full k-NN accuracy requires rebuilding per-photo embeddings locally (see `docs/dataset.md`).

2.3 Data (what exists vs what we release)

Layer	Production	This repo
----- ----- -----		
Photographs	~1.06M on disk (Minka, iNaturalist, GBIF, ...)	Not released (licence)
Per-photo embeddings	1,118,353 vectors	Not released (size + derived from photos)
Prototype per species	4,543 × 1024 float32	Released (~15 MB)
Taxon IDs / names	2,985 catalog rows	Released (dataset/catalog.json)
Calibrator, geo priors, cryptic pairs, exceptions	live JSON	Released

Names are cross-checked with **WoRMS** (accepted name stored beside the stable slug). Images are filtered for format/resolution; field-guide plate crops that mixed several taxa under one label were quarantined (1,088 photos / 107 species in the August 2026 audit).

Tiers (catalog, not gallery size):

Tier	Meaning	n taxa
----- ----- -----		
0	Heterobranchs (expert collaborators' core group)	634
1	Other marine	1,527
2	Terrestrial / incidental in marine photos	824

3. Evaluation protocol

Headline numbers use held-out **observations** (an encounter, often several photos of one organism), not a random photo split. Photo-level 80/20 inflates k-NN and metric-learning numbers by ~10 pp because citizen-science bursts are near-duplicates.

Rule	Why
----- -----	
Stratify by observation ID	No burst leakage
Reject a candidate if it is too similar to the live gallery	Catches missing IDs after path migrations
Same protocol across ablations	Comparable McNemar

Do not mix harvests when citing a delta	Different species mix $\neq$ model gain
---	---

Incident (2026-09-23). The August gate below was itself ineffective: it compared a TTA-averaged (orig + 90% crop) query embedding with gallery embeddings at cosine >0.999 ; an identical photo scores ~ 0.99 that way. 50.8% of the evaluation set had re-entered the gallery. The gate now compares the **global** embedding (as stored in the gallery) at ≥ 0.98 , and an audit script re-checks every evaluation row after any gallery growth (O18).

Incident (2026-08-25/26). An observation-ID denylist pointed at an abandoned path and silently matched nothing. 42.7% of a 22,332-photo harvest were already in the gallery. At $k=15$ the headline only moved ~ 1 pp; at small k the curve was absurd (accuracy rising toward $k=1$). Fix: embed each candidate and drop near-duplicates. Kept as a standing harvest gate.

Two labelled metrics in this paper

Label	Cohort	Species top-1	Use
----- ----- ----- -----			
A — TTA-era freeze	harvest_calib n=12,788 (Aug 2026, leak-checked)	75.97% (later 77.77% with inference stack; 79.10% jsonl after densification)	Comparable ablations in §4.1–4.2
B — live calibrator	calib_raw_t05 n=19,087 (2026-09-14)	85.78%	Current production report
C — leak-free	calib_raw_t05 n=12,373 (2026-09-23, audited, ≥ 0.98 copies removed)	88.11%	Current production report (§5)

B is larger, covers more hard/rare species, and uses $T=0.05$ + later gallery quality work. **A is not “worse engineering”**; **B is not a 10-point magic leap on the same photos.**

4. Experiments

Each row: **hypothesis** → **result** → **contribution to the project**. Details and McNemar counts: docs/EXPERIMENTS.md.

4.1 Kept (in production)

ID	What we tried	Result	What it contributed
---- ----- ----- -----			
K1	Replace ViT-L gallery with frozen ViT-H	+6.8 pp species (63.9% → 70.6% on the then cohort)	Largest single modelling gain. Production encoder.
K2	k-NN grid, observation-stratified	k=15 beats $k=8$ and $k\geq 20$	Production k.
K3	Unify SSD + HDD-archive photos and re-embed	71.7% → 75.8% on leak-checked n=12,788	Closed the archive gap (see docs/archive/ARCHIVE_GAP.md). Completeness > adapters.
K4	Logistic calibration on 10 k-NN features	ECE ~ 0.03 vs ~ 0.09 raw cosine	Interpretable AutoID threshold.
K5	Hierarchical abstention (margin + expert pairs/genera)	Genus/family fallbacks $\sim 89\%$ when used; species top-1 slightly conservative	Safer citizen-science output.

K6	90% centre-crop TTA, then 65% ROI fusion	TTA +0.21–0.75 pp; ROI fusion +1.63 pp vs global-only on n=12,788	Production query embedding.
K7	Late fusion of multiple photos in one observation	+0.79 pp full set; +3.15 pp on the 25% multi-photo subset	Production when N>1.
K8	Fisher / PCA+LDA local subspace on documented cryptic pairs	+0.14 to +0.36 pp on the official harvest	Scoped re-rank; not a new encoder.
K9	Inter-genus same-family local subspace	Official re-harvest 77.77% (cohort A + inference stack)	Production second trigger.
K10	Gallery densification (more/better reference photos, frozen encoder)	Cohort A jsonl 79.10% ; later quality swaps (curators, Q≥8, seagrass)	Data, not weights. Live gallery 848,883.
K11	Tempered k-NN aggregator T=0.05	McNemar n=8,000: +7.81 pp , 671/46	Production scorer (cumulative with K10).
K12	MiniCPM gate on two confusion pairs (Branchiommia/Myxicola, Cerianthus/Pachycerianthus)	Small-n McNemar 12/1, p=0.003	Gated sidecar, not a general VLM re-ranker.
K13	Nomenclature audit (WoRMS + GBIF)	85 accepted-name fields; 4 duplicate-slug merges	Catalog hygiene; slugs never used as WoRMS names.
K14	Expert indistinguishability list + sponge group abstain	Extra pairs from eval confusion (≥33% reciprocal)	Decision layer where vision saturates.
K19	Grow reference photos to ~1,000/species where the sources have them + full per-species re-embed	McNemar n=18,000: +0.57 pp (313/211, p=1e-5). By species: grew ≥+300 vectors +7.7 pp , +100–299 +12.0 pp , +1–99 +3.4 pp; species that did not grow –0.63 pp (dilution)	Data works where it exists; 77% of remaining errors sit in species that did not change.
K20	Second growth wave (121 species, 21.9k photos), rebuild to 1,118,353 vectors	McNemar n=21,000: +0.22 pp (84/38, p=5e-5); touched species 83.3% → 89.5% (83/1); 31 improve, 0 worsen	In production 2026-09-21.
K21	Independent second opinion for the closed-set retriever: BioCLIP zero-shot over a regional checklist	300 Mediterranean bird observations: BF≥0.85 and zero-shot≥0.80 agree → 98.5% precision at 69% coverage (BF alone 91.2% at 79%). All groups: 95.9% at 49% with a rescue tier vs BF alone 93.8% at 48%	Open-set guard for AutoID (birds in production; other groups pending).
K23	Per-species vote cap in the k-NN aggregator (KNN_CLASS_CAP=3)	McNemar n=12,000 held-out obs: +1.13 pp (92.33% → 93.47%; 228 fixes / 92 breaks); 177 species improve / 70 worsen ; gains largest for species with <25 gallery photos (+3.2–3.5 pp). cap1 –0.19, cap2 +0.82, cap4 +1.08, cap5 +0.94 pp	In production 2026-09-22 07:17 CEST. (Measured before the O18 purge; the relative gain is what counts.)
K22	Promotion criterion counted in species	173 species improve / 153 worsen / 2,186 unchanged (previous → current index)	Report the per-species balance next to Δ and p.

4.2 Rejected (do not repeat as-is)

ID	What we tried	Result	What it contributed
--- ----- ----- -----			
R1	QLoRA ViT-L, trainable projection	1.7% species	Compatibility/fragility lesson; not a BioCLIP failure in general.

R2	QLoRA BioCLIP-2 ViT-L vs ViT-H prod	Architecture mismatch (768 vs 1024-d)	Discarded before eval.
R3	Triplet on ViT-H (8 variants)	-0.7 to -7 pp	Metric learning on this gallery memorizes.
R4	ArcFace on frozen ViT-H	Tie with k-NN (-71.4 vs 71.6)	No reason to replace retrieval.
R5	LoRA+ArcFace 100 spp	+0.0 pp OOS	Internal splits lied.
R6	LoRA last-4-blocks + ArcFace, 1,358 spp	-31.2 pp on full eval	Fine-tuning a subset poisons the shared space.
R7	Frozen-backbone linear head	-0.6 to -1.1 pp OOS (train mini-set had shown +2.6)	Train/eval disagreement; no cutover.
R8	Weight expert-guide crops in the gallery	-0.9 pp	OCR plates ≠ underwater photos.
R9	Near-duplicate burst removal (cos>0.99)	-1.6 pp	Bursts help k-NN.
R10	Outlier filtering on embeddings	-0.21 to -1.45 pp	Removed intra-species variation.
R11	Widen same-genus abstention margin	9:1 cost of true species IDs vs recovered errors	Production τ stays tight.
R12	“Prefer epibiont” re-rank	-0.23 pp (64 fixed / 116 broken)	Host photos look like the partner in top-k.
R13	Scoped SupCon re-ranker on 20 cryptic pairs	Val loss diverged from epoch 1; kill-switch before the eval set	Third independent architecture, same failure mode.
R14	Horizontal-flip TTA	-0.13 pp	Asymmetry is signal.
R15	Per-species prototype weight by dispersion	-0.09 pp	Global dispersion ≠ this query.
R16	Family-consensus penalty on k-NN	-0.41 pp, p=0.0005	Majority vote is already wrong on convergent morphologies.
R17	Attention-guided crop (Bucket B)	+0.04 pp, p=0.77	Noise.
R18	Phylum/class consensus filter	All thresholds net-negative	80% of cross-group errors are already in k-NN.
R19	Ancestral subspace for spp with <5 refs	1 addressable eval observation	Closed before GPU spend.
R20	Background/substrate neutralization	-0.92 pp vs own control; -3.48 vs full stack	Habitat is signal for epibionts.
R21	Extra geo prior from Minka point clouds on top of production geo	Not significant; 4/5 OOF folds chose strength 0	Confusable species co-occur.
R22	DINOv2 fusion (scripts misnamed “DINOv3”)	-6.75 pp vs production	Prototypes-only DINOv2 is far weaker here.
R23	Morphological multi-prototypes (k-means)	Not significant; 4/5 folds blend=0	Single centroid already sufficient for the boost term.
R24	Selective marine re-embed (36 spp)	~0 pp	No cutover.
R25	Seagrass VLM / generic densification	Null	Quality swap helped <i>Posidonia</i> , not <i>Cymodocea</i> .
R26	MiniCPM “pure subject” filter on fauna (16 spp)	97.5% already “pure”	Does not transfer from seagrass.
R27	Broad Q≥8 photo swap on already-exhausted taxa	Very low yield on some batches; high yield on others	Quality helps when better photos exist; not a universal lever.

R28	Raise the confidence threshold to cut confident errors	Live backtest n=184: precision flat at 90–92% for $p \geq 0.80 \dots 0.95$; only coverage falls (52% → 11%)	Calibration saturates; not a lever.
R29	Require iNaturalist CV agreement before publishing	Of 81 BF answers ≥ 0.85 : iNat agreed on 51 (BF right 96%), returned nothing on 29 (BF right 83%), disagreed on 1; ~4 correct answers blocked per error avoided	Replaced by domain guards + zero-shot consensus (1-day trial, audit pending).

4.3 Operational failures (not model ideas, still public)

These are included because they change how numbers should be read.

ID	Failure	Effect	Contribution
----	---------	--------	--------------

-----|-----|-----|

O1	Calibration harvest leaked into the gallery (Aug 2026)	Inflated low-k curves	Similarity dedup is mandatory.
O2	FAISS index vs <code>species_ids</code> desync after <code>/reload</code> (1 Sep 2026)	Live IDs assigned terrestrial queries marine names at 85–100% confidence; disk evals valid	Reload index and labels together.
O3	Admin metrics reading frozen snapshot files	Panel showed stale/in-sample accuracy	Live metrics from the current jsonl.
O4	Empty <code>--species</code> list after nested SSH (13 Sep 2026)	Quality-swap ran on extra taxa for ~35 min	Pass slug files, never interpolated lists.
O12	The serving process translated FAISS labels with the live species folder list, not the index's own name list; three new folders shifted every label while <code>/health</code> still reported aligned	Wrong species at 0.92 similarity for hours (2026-09-20/21); AutoID published nothing	Guard cron compares index names ↔ loaded species ↔ folders (extends O2).
O13	An unattended job rebuilt the production FAISS directory and refit calibration before validation finished	Production index overwritten by an unvalidated build	Unattended jobs never write production paths.
O14	The Minka web host returns 403 to the default <code>python-httpx</code> User-Agent (login and photo downloads)	AutoID zero publications for days (second recurrence of the missing-header class)	One shared HTTP client; alarm on zero-publication days.
O15	Two agent sessions ran the same cycle in parallel; one promotion overwrote the other; GPU OOM with sidecar + serving + re-embed	Duplicated compute; index/calibration mismatch for ~30 min	Single active session with a written registry.
O16	Closed-set, observer-correlated evaluation: panel 92.4% vs ~80% on 300 recent research-grade observations; 12% of bird observations are species outside the catalog	Panel overstates real-world accuracy	Recent-observation sample as operating KPI.
O18	Evaluation leak gate compared a TTA-averaged embedding with a 0.999 threshold; identical photos score ~0.99, so the gate never fired. Later ad-hoc searches (Wikimedia/GBIF/iNat any grade) had no gate at all	50.8% of evaluation rows were gallery copies; panel 92.4–93.4% vs 88.1% clean; rare-species accuracy on unseen photos 24% hidden as 57%	Global-embedding gate ≥ 0.98 in every harvester; <code>leak_audit_calib.py</code> after every gallery growth; leaked rows kept aside, calibrator refit (2026-09-23).

O17	Dashboard accuracy per species changes only when the eval is re-scored against the serving index	Numbers frozen after promotions	Daily automated re-score; re-score after every promotion.
-----	--	---------------------------------	---

5. Current production report (cohort C, leak-free, 2026-09-23)

Level	Accuracy	n
-------	----------	---

|-----|-----|---

Species	88.11%	12,373
Genus	90.55%	12,373
Family	92.46%	12,373
Tier 0 (heterobranchs)	85.12%	2,446
Tier 1 (other marine)	87.72%	5,633
Tier 2 (terrestrial/incidental)	90.34%	4,294

Per species (2,091 with clean evaluation): 1,429 at 100%, 431 below 80%. 894 catalog species lost their only evaluation rows in the purge and are being re-harvested with the fixed gate. Cohort B (85.78%, n=19,087, 14 Sep) is kept for history; it was affected by the same leak.

Remaining error is dominated by **visual cripsis and morphological convergence** (sponges, filamentous algae, some fishes, seagrass blades), not by “we need LoRA”. Coverage: most catalog species now have ≥ 1 held-out observation; a small tail is exhausted on Minka+iNaturalist (dozens of taxa with essentially no extra research-grade photos).

Seagrass (n=60 each, 14 Sep): *Posidonia oceanica* 73.3%; *Cymodocea nodosa* 70.0% (quality swap **not** shipped — confusion with *Nanozostera*); *Nanozostera noltii* 86.7%; *Zostera marina* 75.0%. The pair *Z. marina* $\leftrightarrow$ *N. noltii* was added to expert abstention after 20/120 cross-confusions.

6. Limitations

1. Species top-1 is not expert-level on cryptic invertebrates.
2. No geographic generalization study outside the Mediterranean.
3. Public code is a **reconstruction identifier** (prototypes + optional local k-NN), not a dump of HanSolo’s 1,300-line service (sidecars, AutoID, GPU contention with MiniCPM).
4. AutoID precision/coverage is measured on a species-disjoint split of the leak-free set (§2.2); it is closed-set and has not yet been audited against community identifications.
5. Curator corrections are not yet a closed training loop.
6. **Closed-set retrieval:** species absent from the catalog get a confident wrong answer; the calibrator only saw catalog species. Mitigated for birds by a zero-shot consensus guard (K21); not yet for other groups.
7. **Evaluation optimism:** the leak-free panel metric (88.1%) is still closed-set and observer-correlated; a recent-observation sample (~80%) is the operating KPI (O16). Until 2026-09-23 the panel also contained gallery copies (O18).

8. **AutoID without iNaturalist verification** has run for one day only (guards: Mediterranean bounding box, bird consensus, hourly cap, now 30/h and 1,000/day; reaching the hourly cap pauses that hour instead of tripping the circuit breaker); the audit against community identifications is pending.

7. Reproducibility

```
` bash

git clone https://github.com/yespi/biofauna.git && cd biofauna

pip install -r requirements.txt

python -m uvicorn src.identify_service:app --host 0.0.0.0 --port 8090
```

```
`
```

Downloads BioCLIP-2.5 from Hugging Face on first run. Uses `data/patterns/` prototypes. To rebuild the full gallery: download photos with taxon IDs in `dataset/catalog.json`, run `scripts/reembed_vith.py`, place `embeddings.npy` beside each prototype.

Licence: MIT for code and released JSON/prototypes. Photograph copyright remains with original observers and platforms.

Acknowledgments

Xavier Salvador, Miquel Pontes, and Manuel Ballesteros (GROC/OPK, published checklists). Minka and iNaturalist observers. WoRMS. BioCLIP (Stevens et al.).

Data availability

Code, prototypes, catalog, calibrators, geo priors, cryptic pairs: this repository. Backbone: Hugging Face `imageomics/bioclclip-2.5-vith14`. Images: obtain independently from Minka/iNaturalist/GBIF.

References

1. Stevens, S. et al. (2024). BioCLIP: A Vision-Language Model for the Tree of Life. *CVPR*.
2. Dettmers, T. et al. (2023). QLoRA. *NeurIPS*.
3. Hu, E.J. et al. (2021). LoRA. *ICLR*.
4. Radford, A. et al. (2021). CLIP. *ICML*.
5. Van Horn, G. et al. (2018, 2021). iNaturalist datasets. *CVPR*.
6. Khosla, P. et al. (2020). Supervised Contrastive Learning. *NeurIPS*.
7. Coll, M. et al. (2010). Mediterranean marine biodiversity. *PLoS ONE*.

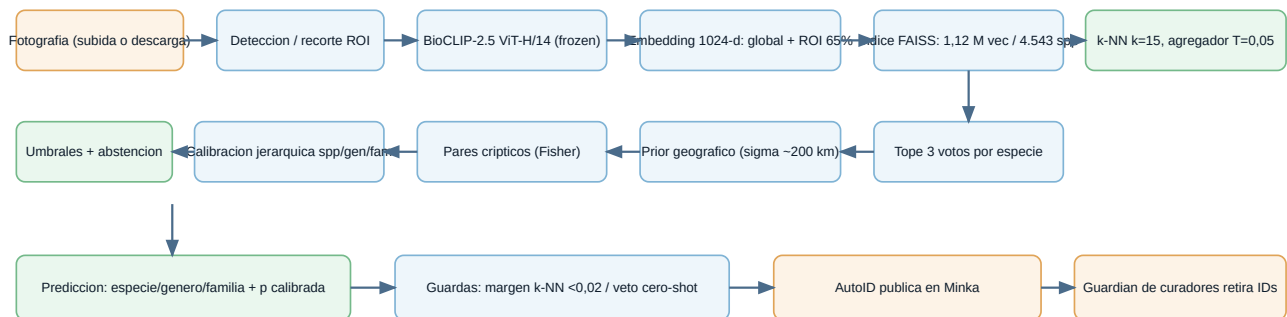
8. Bianchi, C.N. & Morri, C. (2000). Marine biodiversity of the Mediterranean Sea. *Mar. Pollut. Bull.*
9. Ballesteros, M. (2007). Opisthobranchs of the Catalan coasts. *SPIRA*.
10. Cervera, J.L. et al. (2004). Iberian opisthobranch checklist. *Bol. Inst. Esp. Oceanogr.*
11. Salvador, X., Lázaro, J. & Fuentes, M.A. (2022). Invertebrats marins de la Vall del Ridaura. Digital CSIC.
12. Schroff, F. et al. (2015). FaceNet. *CVPR*.
13. Hermans, A. et al. (2017). Triplet loss. arXiv:1703.07737.

Technical report accompanying the open repository. Not a journal submission. Target venues after a frozen PDF cut: Ecological Informatics / Biodiversity Data Journal / PeerJ.

7. How BioFauna identifies a species

1 - Como BioFauna identifica una especie

Cada paso es auditable; k, T, cap y umbrales son los de producción.



Rescate cero-shot (2026): si la similitud top-1 < 0,85 y el cero-shot del catálogo (2.985 etiquetas) coincide con otra especie con $p \geq 0,90$, gana el cero-shot: +0,91 pp medidos en el eval de campo.

Control de fugas: ninguna foto de evaluación puede estar en el índice (auditoria global $\geq 0,98$; purga automática).

La ID de un curador manda siempre; el modelo se audita con sus correcciones.

Pipeline: from a photograph to a calibrated, guarded answer

BioFauna is a **retrieval** classifier, not a trained end-to-end network:

1. **Frozen encoder.** Every gallery photograph is embedded with **BioCLIP-2.5 ViT-H/14** (frozen, no fine-tuning). The query image is embedded the same way, as the global vector ****plus a 65 % region-of-interest fusion**** (the subject-centred crop), L2-normalised.
2. **Approximate search.** A **FAISS** index holds all gallery vectors (currently ****1.12 M vectors / 4,543 species****, row-aligned so that each vector maps to exactly one species).
3. **k-NN with a tempered aggregator.** The $k=15$ neighbours vote with weights $\exp(\text{similarity} / T)$, $T=0.05$, **capped at 3 votes per species** so that a single over-represented species cannot swamp the vote.
4. **Context corrections.** A **geographic prior** (1,386 species, $\sigma \approx 200$ km) and **cryptic-pair** handling

(Fisher directions / local subspace) correct the ranking when two species are visually almost identical.

5. **Hierarchical calibration.** A per-species logistic model converts the k-NN score into a ****calibrated probability, with genus- and family-level fallbacks; below the calibrated threshold the service abstains**** to genus/family instead of guessing.

6. **Zero-shot rescue (2026).** When the top-1 similarity is low (< 0.85), a **catalogue-wide zero-shot** comparison (2,985 text labels) is consulted; if it agrees with a *different* species at $p \geq 0.90$, it wins.

Measured effect: **+0.91 pp** on the field evaluation.

7. **Publication guards (AutoID).** A species is not published when the top-1/top-2 similarity **margin < 0.02**, when the observation falls outside the domain, or when a vocal disagreement exists; a **curator guard** automatically retracts our identifications when a curator corrects at class level or above.

8. How the model improved, step by step



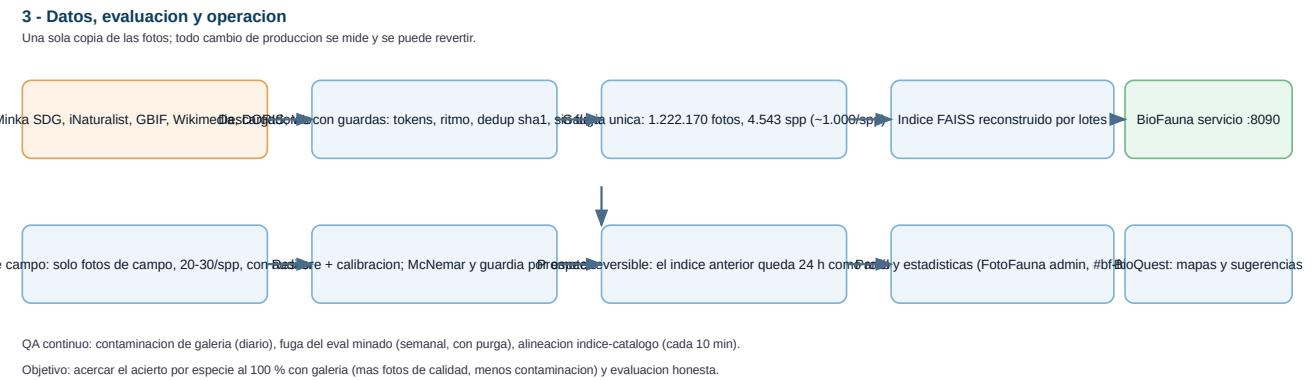
Improvement timeline

step	change	effect
--- --- ---		
YOLOFauna (2024–2026)	BioCLIP ViT-L + k-NN	63.9 %
Aug 2026	full re-embedding with BioCLIP-2.5 ViT-H/14	70.6 %
Aug 2026	k=15 + hierarchical fallback + calibration hygiene	71.7 %
Aug 2026	hierarchical calibration (species/genus/family)	74.1 %
Sep 2026	3-vote cap per species, ROI fusion, geographic prior	77.9 %
Sep 2026	evaluation leak fixed → honest panel (near-100 % of the drop was leakage, not error)	79.4 %
Sep 2026	gallery hygiene: 135 duplicate synonym slugs merged (Minka authority), ~1,000 mislabelled photos relocated, 57 quarantined	79.6 %

Sep 2026	zero-shot rescue + publication guards	79.6 % (+0.91 pp on the same rows; the headline moves with the evaluation)
----------	---------------------------------------	--

Two lessons that shaped the project: (a) **the evaluation is part of the model** — an unfixed leak in the evaluation made the system look better than it was; (b) **gallery quality beats model complexity** — synonyms, contamination and mislabelled observations cost more accuracy than any hyper-parameter we tuned (LoRA/QLoRA/triplet/ArcFace/SupCon on this embedding space were all rejected by measurement).

9. Data, evaluation and operation



Data flow and operations

- **Sources** (public images): **Minka SDG** 377,476 · **iNaturalist** 779,411 · GBIF 39,570 · Wikimedia Commons 6,677 · DORIS/FFESSM 5,091 · SeaSlugForum 3,147 · WoRMS 1,320 · FishBase 1,072 → **≈1.22 M photographs.**
- **One copy of each photo**, never deleted; every relocation is done with a manifest and a rollback script.
- **Field evaluation:** field-only photographs (iNaturalist/Minka research grade), 20–30 per species, leak-audited before merging (a gallery copy in the evaluation would inflate the score).
- **Every production change is measured** (McNemar + per-species guard); the previous index stays as a **24-h rollback.**
- **Continuous QA:** gallery contamination (prototype-centroid metric, daily), mined-evaluation leak audit (weekly, with automatic purge), index ↔ catalogue alignment (every 10 min).